

Toward A Data-driven Cyberinfrastructure Tool for Urban Micromobility-Friendly Path Planning: A Hartford Study

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Abstract

The increasing demand for sustainable and affordable urban mobility and connectivity has underscored the need for cohesive, micromobility-friendly paths as well as micromobility networks, serving both cyclists and electric (e)-scooter riders. To this end, this paper aims to explore and investigate how to (1) evaluate the micromobility network connectivity as well as their friendliness and stress for riding micromobility vehicles; and (2) recommend connection of fragmented micromobility networks and their upgrades within a city. To this end, we present a Micromobility infrastructure Planner for Low-stress and Accessible Networks (M-PLAN), a geospatial cyberinfrastructure tool for planning support of micromobility-friendly, user-centered networks. M-PLAN analyzes OpenStreetMap (OSM) as well as Hartford's Bicycle Master Plan data to identify existing micromobility infrastructures, determines micromobility stress levels based on geospatial attributes, and recommends targeted upgrades and network expansions. Our tool applies a Minimum Spanning Tree (MST) algorithm to expand networks, and integrates Dijkstra's algorithm to evaluate low-stress connectivity to destinations. Our real-world case studies in the City of Hartford, Connecticut (CT) have demonstrated the M-PLAN's effectiveness in identifying infrastructure gaps and recommending improvements for low-stress, connected micromobility networks. The M-PLAN cyberinfrastructure project will have broad impacts in serving the geospatial research communities and beyond (**Project Website**: <https://github.com/annette-grabowski/M-PLAN-SIGSPATIAL-26>).

CCS Concepts

• **Information systems** → *Decision support systems*; • **Computing methodologies** → *Artificial intelligence*.

Keywords

Cyberinfrastructure, Micromobility

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1 Introduction

The rising demand for sustainable urban transportation has underscored the need for effective, data-driven mobility infrastructure planning tools [5, 12]. In many U.S. cities with fragmented micromobility networks and emerging active transportation needs, there arises an urgent need for practical methods to evaluate existing conditions and recommend strategies to expand and connect these networks in ways that improve safety, connectivity, and usability for micromobility users (e.g., cyclists, e-scooter riders, wheelchair users, delivery robots) [8].

To this end, we present a data-driven cyberinfrastructure tool named M-PLAN (Micromobility infrastructure Planner for Low-stress and Accessible Networks) to assist micromobility-friendly, user-centered planning of low-stress and accessible networks. We aim to design a cyberinfrastructure tool to support our research community and city planners to understand and plan the micromobility-friendly, user-centered path based on OpenStreetMap (OSM) data [7] as well as other exogenous data sources. This way, we can help identify and quantify the micromobility stress and friendliness for the riders, and lay out the tool designs toward effective decision-support of meaningful micromobility network upgrades.

Based on our extensive, local community engagement at the City of Hartford, Connecticut (CT), the objective of our M-PLAN research is to (1) outline the framework design on how to evaluate the existing existing micromobility networks (e.g., bike lanes) connectivity and stress; and (2) explore and understand the opportunities on how to provide decision-support on connecting and linking the fragmented networks for the local cyclists and e-scooter riders. We will particularly answer the following two research questions through this Hartford study.

Q1 – How can we incorporate multi-source data to assess the micromobility-friendliness in urban micromobility networks? Based on our extensive micromobility studies (supported by the Capitol Region Council of Governments (CRCOG) in Hartford) in the Greater Hartford Metropolitan Area since 2022, we have observed that urban micromobility networks often “grow” in an ad hoc and disjoint manner, resulting a largely fragmented network structure that limit the connectivity and adoption. To address this, we envision that effective planning recommendations must account for such key variables as infrastructure type, road classification, speed limits, lane counts, on-street parking, and surrounding land-use zoning. There arises a need from the local city planners of a data-driven cyberinfrastructure tool to comprehensively take in multi-source data, evaluate existing micromobility networks, and assess the possible expansion opportunities.

Q2 – How can we provide usable decision-support in refining the urban micromobility networks? The usability of

the micromobility network planning decision-support resides in the micromobility-friendliness enabled by the cyberinfrastructure. We note that a micromobility-friendly, user-centered path should account for reducing the stress level — including physical and cognitive safety burden — potentially experienced by the micromobility riders. Such micromobility stress is deeply associated with the built environment. A cyclist exposed to, for instance, fast or heavy motor-vehicle traffic, close passing, and lack of separation tends to feel more micromobility stress than they are along a protected bike lane. We also observe that the stress resides in the perceived danger, discomfort, and lack of control in human-mobility interaction — that is, when the riders are interacting with the built environments. These require quantitative and qualitative characterization of micromobility networks in order to provide usable decision support of the micromobility communities and city planners.

To explore answers to these questions, we have conducted a case study in the City of Hartford, CT. We have made the following contributions in designing, implementing, and evaluating M-PLAN.

- M-PLAN takes in the open-source OpenStreetMap (OSM) as well as Hartford’s Bicycle Master Plan [1] data created by the City of Hartford in order to cross-reference and classify the existing micromobility network paths and quantify their micromobility stress levels. We have designed a network expansion approach based on the minimum spanning tree (MST) to connect and expand from the fragmented networks, and evaluated the low-stress connectivity toward the key locations as destinations inside the city for potential network upgrades.
- Our initial experimental results demonstrate that M-PLAN is effective in identifying infrastructure gaps, evaluating the overall connectivity and usability, and recommending improvements of urban micromobility networks. We have integrated the Google Maps Street View Static APIs to further examine M-PLAN’s micromobility stress classifications. Our case studies have highlighted the key infrastructure gaps, and offered actionable recommendations for improving network connectivity in the City of Hartford.

The rest of this short paper is organized as follows. Section 2 first reviews the related work and prior studies that have inspired the development of M-PLAN. Section 3 presents the core design and algorithmic implementation of M-PLAN. Section 4 presents the case study and results from the City of Hartford, CT. We conclude this short paper and discuss our future work in Section 5.

2 Related Work

We review the related studies and present differences of M-PLAN.

BikeDNA [11] is an open-source Python package designed to evaluate the completeness and quality of bicycle infrastructure data. BikeDNA provides the infrastructure classification in terms of protected and unprotected bike lanes, and assess the network topology. Despite their effectiveness in intrinsic data consistency check and extrinsic data cross-referencing, the package has not accounted for and evaluated the impacts of micromobility stress, and thus cannot provide more micromobility-friendly, user-centered network expansion decision support.

The BNA (Bike Network Analysis) tool [10] studied traffic stress modeling to assess low-stress connectivity to destinations. The tool

integrates data from OSM, census blocks, and municipal boundaries, and classifies routes based on factors such as road width, number of lanes, and presence of bike lanes. However, the BNA tool does not enable decision-support in suggesting infrastructure upgrades (e.g., connection or expansion) or handling data gaps effectively, and relies on default assumptions for handling missing data.

The Indianapolis Metropolitan Planning Organization (MPO) developed a Bike Network Analysis StoryMap [4] to visualize network stress scores and identify strategic connection points between low-stress segments. Similar to the BNA tool, the StoryMap emphasizes the local context and visualization to highlight potential micromobility infrastructure gaps. However, it does not incorporate real-world validation or offer automated recommendations for infrastructure growth.

Inspired by the above tools, our M-PLAN advances from the following perspectives. M-PLAN jointly takes into account OpenStreetMap and Hartford’s Bicycle Master Plan as the open and local data sources for micromobility network data analysis. Our tool has offered a reproducible algorithmic method to expand low-stress networks and optimize the micromobility-friendly routes. We have further designed and integrated the Google Street View APIs for validating the micromobility stresses of the streets and providing more human-aligned, user-centered decision support.

3 Core Design of M-PLAN

We outline in this section the designs and implementation of M-PLAN, which evaluates and expands urban micromobility networks by integrating micromobility stress modeling, network expansion, and street view examination. M-PLAN generates actionable insights for our target stakeholders — such as city planners and geospatial researchers — to improve the urban micromobility network connectivity, safety, and planning efficiency. We show in Figure 1 all the key components inside M-PLAN.

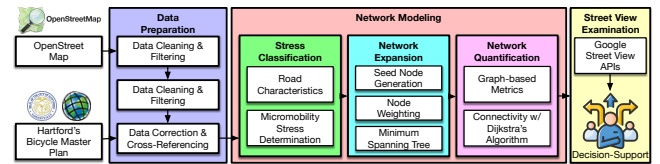


Figure 1: Workflow overview of our M-PLAN prototype, which consists of data preparation, network modeling, and street view examination.

3.1 M-PLAN Data Preparation

The basic workflow of M-PLAN is as follows. M-PLAN begins by downloading OpenStreetMap (OSM) data for a user-defined study area, specified via a geospatial polygon. M-PLAN filters out irrelevant road types such as motorways and private drives, and then processes relevant OSM tags to identify existing bicycle infrastructure. Here our micromobility-relevant infrastructures are categorized into four types: (1) sharrow (a pavement marking painted on roads to indicate that the lane is shared by bicycles and motor vehicles), (2) painted lane (standard bicycle lane), (3) buffered lane (upgrade to painted lane with extra painted buffer zones), and (4) protected lane (separated bike lane).

To ensure consistency and completeness, M-PLAN compensates for discrepancies in how bicycle infrastructure is tagged across



Figure 2: Data preparation: (a) infrastructure type; (b) speed limits; (c) numbers of lanes; and (d) parking availability.

different OSM layers. This is achieved by matching the bicycle network data with the corresponding motor vehicle drive network, helping to fill gaps and create a more accurate representation of the city’s infrastructure. We note that compared to European cities like Copenhagen, Hartford’s infrastructure tagging in OSM was sparse — less than 10% of roads included key attributes such as speed limits, road width, surface type, or lighting conditions. Such lack of coverage suggests that Hartford’s micromobility network remains both underdeveloped and difficult to assess using OSM alone. This motivates our integration of Hartford’s Bike Master Plan data as well as Google Street View examination for cross validation. In particular, we observed that only 59% of Hartford’s municipal bike network segments from the Master Plan matched the OSM data, while 85% of OSM segments matched the Master Plan. This asymmetry indicates that the municipal dataset of Hartford’s Bike Master Plan is more complete and better reflects existing infrastructure. We note from our practices that when key OSM tags are missing, M-PLAN applies default settings based on road classification, which can be further customized by our city planners and geospatial researchers. For instance, primary roads are considered to have a speed limit of 40mph as well as two lanes per direction unless otherwise specified. M-PLAN users can adjust their default settings related to on-street parking (e.g., presence or absence of parking lanes) to reflect local conditions.

3.2 Network Modeling in M-PLAN

We present M-PLAN’s micromobility network modeling in terms of (1) Micromobility Stress Classification, (2) Micromobility Network Expansion, and (3) Network Quantification as follows.

(1) Stress Classification. A key feature of M-PLAN is its ability to model micromobility stress levels for all streets. The micromobility stress is determined based on road characteristics such as type, speed limit, lane count, the presence of on-street parking, and the presence of bicycle infrastructure, as illustrated in Figure 2. M-PLAN first classifies each road segment’s micromobility stress level (high or low), and then identifies the minimum infrastructure upgrade needed to reduce stress to a “low” level, improving safety and usability for the micromobility users. The stress assignment logic, which is based on a set of classification rules defined by road characteristics (such as road infrastructure type, parking availability, and highway type; extensible given additional data inputs), is summarized in Algorithm 1 and provides the decision-support of upgrades for each street segment.

(2) Network Expansion. To expand the network, M-PLAN uses a Minimum Spanning Tree (MST) algorithm. In our Hartford’s studies, seed points are spaced 1,707 meters apart, and snapped to either existing infrastructure or the centroids of land-use zones within

Algorithm 1 Micromobility Path Stress Classification

Input: Number of lanes per direction L , bike infrastructure type B , speed V (unit: mph), parking availability P , highway type H
Output: Path stress level $S \in \{\text{Low, High}\}$

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1: if  $B = \text{PROTECTED}$  then           ▶ Protected facilities are treated as low-stress paths.
2:    $S \leftarrow \text{Low}$ 
3: else if  $H = \text{LIVINGSTREET}$  then       ▶ Living streets are considered to have low
   vehicle speed and few conflict.
4:    $S \leftarrow \text{Low}$ 
5: else if  $L = 2$  then                   ▶ Evaluate roads with two lanes per direction.
6:   if  $B = \text{PAINTED}$  and  $V \leq 20$  then ▶ Painted bike lane is acceptable only at
   low speed.
7:      $S \leftarrow \text{Low}$ 
8:   else if  $B = \text{BUFFERED}$  and  $V \leq 30$  and  $P = \text{No}$  then ▶ Buffered lane
   remains low stress when parking conflict is absent.
9:      $S \leftarrow \text{Low}$ 
10:  else
11:     $S \leftarrow \text{HIGH}$                  ▶ Otherwise, two-lane roads are high stress.
12:  end if
13: else if  $L = 1$  then                   ▶ Evaluate roads with one lane per direction.
14:   if  $V \leq 20$  then                   ▶ Very low-speed roads are considered low stress.
15:      $S \leftarrow \text{Low}$ 
16:   else if  $V \leq 25$  and  $H = \text{RESIDENTIAL}$  then ▶ Residential context lowers
   stress at moderate speed.
17:      $S \leftarrow \text{Low}$ 
18:   else if  $B = \text{PAINTED}$  and  $V \leq 25$  then ▶ Painted bike lane is low stress up to
   25 mph.
19:      $S \leftarrow \text{Low}$ 
20:   else if  $B = \text{PAINTED}$  and  $V \leq 30$  and  $P = \text{No}$  then ▶ No parking reduces
   dooring and curbside conflicts.
21:      $S \leftarrow \text{Low}$ 
22:   else if  $B = \text{BUFFERED}$  and  $V \leq 30$  then ▶ Buffered bike lane supports low
   stress at moderate speed.
23:      $S \leftarrow \text{Low}$ 
24:   else if  $B = \text{BUFFERED}$  and  $V \leq 35$  and  $P = \text{No}$  then ▶ Higher speed is
   acceptable only without parking conflict.
25:      $S \leftarrow \text{Low}$ 
26:   else
27:      $S \leftarrow \text{HIGH}$                  ▶ No low-stress condition is satisfied.
28:   end if
29: else
30:    $S \leftarrow \text{HIGH}$                  ▶ All remaining path conditions are classified as high stress.
31: end if
32: return  $S$                            ▶ Return final stress classification.

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a 500-meter radius. The MST algorithm efficiently connects these points, minimizing infrastructure costs (micromobility stresses) while maximizing connectivity. We note that M-PLAN can optionally weight connections based on land-use type, prioritizing links between residential zones and key destinations [9].

(3) Network Quantification. After generating the proposed network, M-PLAN evaluates its performance using graph-based metrics [11]. These include global efficiency, connected components, and segment lengths. These indices are used to quantify the structure and robustness of the micromobility network, while global efficiency measures the ease of navigating the network from one point to another. These metrics help assess the overall effectiveness of the expanded micromobility network. To assess functional connectivity of the resulting micromobility network, M-PLAN measures destination access using Points of Interest (POIs) such as convenience stores, libraries, and parks. Via the Dijkstra’s algorithm, M-PLAN finds the number of destinations reachable within a 10-minute ride (approximately 2,680 meters) from residential zone centroids in Hartford via our low-stress routes.

3.3 Google Street View Examination

To examine the feasibility of proposed infrastructure upgrades, M-PLAN integrates the Google Maps Street View Static API. For

each sample (street) of proposed expansions or upgrades, M-PLAN identifies the midpoint coordinates of the affected street segment and retrieves a corresponding street-level image. Images in Google Street View are captured with a 0-degree pitch (flat, ground-level perspective) and span a 90-degree segment of the 360-degree Street View panorama. The viewing angle is aligned with the nearest edge in the network to ensure relevance. This visual validation step helps planners assess existing conditions and confirm the practicality of proposed improvements before implementation. We expanded the tool to automatically retrieve Google Street View images for a sample of proposed network upgrades or expansions. For each segment, M-PLAN identifies midpoint coordinates and fetches images using the Google Street View Static API. Images are captured at a 0-degree pitch and aligned with the nearest edge in the network, spanning a 90-degree field of view. This enhancement enables usable, scalable, and real-world validation and supports more informed decision-making by providing visual context for proposed infrastructure improvements.

4 Results from Hartford Case Studies

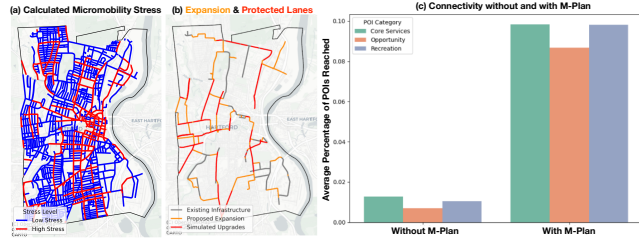


Figure 3: (a) classified micromobility stress, (b) suggested expansion and protected lanes, and (c) quantified connectivity without and with M-PLAN.

We showcase the results in Figure 3. Using M-PLAN, we quantify the stress levels (blue: low-stress; red: high-stress) in Figure 3(a), and generated in Figure 3(b) a network expansion (orange) and protected lane upgrade (red) plan for Hartford. The seed points were placed every 1,707 meters and snapped to existing micromobility infrastructure or land-use centroids. The resulting network prioritized connections between residential zones and key destinations.

We show in Figure 3(c) a bar chart of the average reachable POIs by category for the initial vs. proposed low-stress network and quantify the outcomes of our decision-support. For our initial test of M-PLAN, a single destination type was used for represent each POI category (“core services” like convenience stores, “opportunity” like libraries, and “recreation” like parks). The proposed network by M-PLAN significantly improves low-stress connectivity, reduces fragmentation, and increases overall efficiency. Our preliminary results demonstrate M-PLAN’s potentials to generate practical, data-driven recommendations for infrastructure growth with regard to the study area, infrastructure conditions, and recommended upgrades, helping our city planners and researchers understand gaps, validate assumptions, and prioritize municipality investments in planning networks.

To examine our tool’s stress classification logic, we showcase two cases of streets in the downtown Hartford using the Google Street View screenshots (other samples are left out due to space limit). **(1) Allyn Street.** This is classified as low stress with painted lanes. The street view showed a mix of sharrows and painted lanes,



Figure 4: Two case studies in City of Hartford.

confirming the classification but revealing limitations in M-PLAN, which assigns the highest infrastructure level to the entire segment regardless of variability within it. **(2) Main Street.** This is classified as high stress with painted lanes. The street view revealed buffered lanes and on-street parking, with frequent intersections and dooring risks, confirming the high-stress classification.

5 Conclusion & Future Work

In this study, we developed M-PLAN, a tool for evaluating and expanding urban micromobility networks through data preparation, micromobility stress modeling, and street view examination. Applied to the City of Hartford, CT, we have gained the initial cyberinfrastructure tool design insights regarding how M-PLAN identifies key infrastructure gaps, assesses micromobility stress levels, and generates actionable recommendations for the city planners in improving low-stress connectivities. The M-PLAN cyberinfrastructure project will have broad impacts in serving the geospatial research communities and beyond. Further extension includes integration of geospatial AI techniques such as reinforcement learning [3, 6] for interactive expansion studies [2, 12].

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